VI. CARTESIAN CONTROL OF ROBOT MANIPULATORS

Introduction

Given a starting end effector position and orientation and a goal position and orientation we want to generate a smooth trajectory that takes the end effector from the start to the goal. Since a desired end effector motion is generated by controlling the joint motion, it is necessary to determine the appropriate joint movements that correspond to the desired smooth end effector motion. This can be done by computing, at each instant, the desired end effector velocity, and from that, the desired joint velocities. The latter step involves inverting the manipulator Jacobian matrix. This approach is called the resolved motion rate control (RMRC) scheme and is due to Whitney[1]. The presentation here is a variation of Whitney's scheme. Our geometric approach to constructing the Jacobian in an intermediate reference frame can be used to simplify the computations.

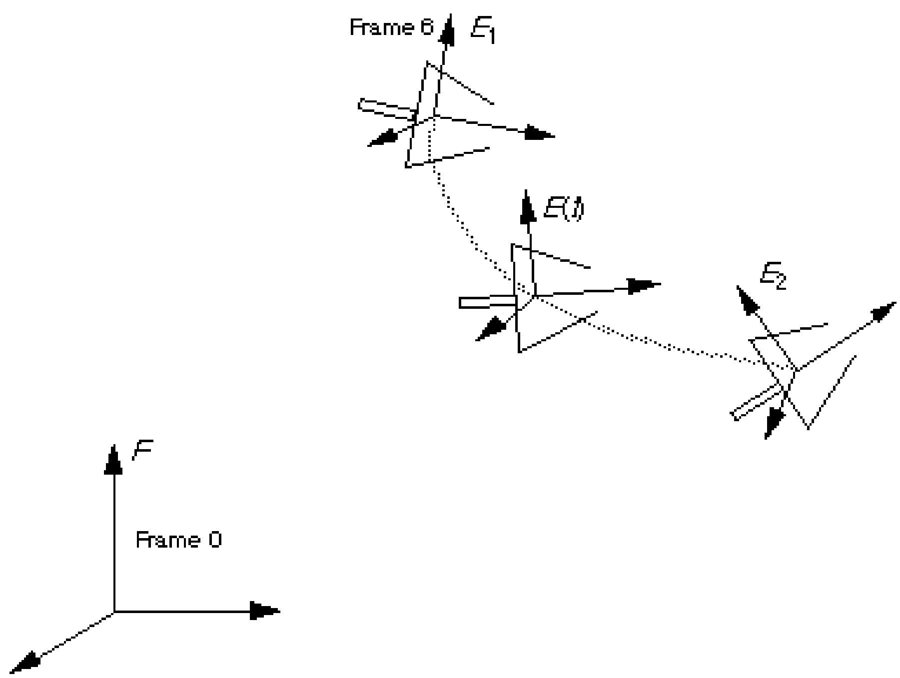

Figure 1: Generation of a smooth trajectory for the end effector

[1]D.E. Whitney. Resolved motion rate control of manipulators and human prosthesis. *IEEE Trans. Man-Machine Systems.* Vol. MMS-10, 1969.

Nomenclature

Let the task be to take the end effector from the starting position and orientation given by frame E_1 to the goal position and orientation given by frame E_2 in τ seconds. The following symbols are used in the development of the RMRC algorithm:

- ${}^0\mathbf{A}_6$ the homogeneous transform that describes the position and orientation of the end effector frame with respect to the base frame
- E_1 the end effector frame (reference frame 6) at the start position
- E_2 the end effector frame (reference frame 6) at the goal position
- F the fixed reference frame (reference frame 0)
- $\mathbf{Q}_i$ the homogeneous transformation matrix for the displacement $D{:}F \rightarrow E_i$ given by

$$\mathbf{Q}_1 = \left[\begin{array}{c|c} \mathbf{R}_1 & \mathbf{d}_1 \\ \hline \mathbf{0} & 1 \end{array}\right], \qquad \mathbf{Q}_2 = \left[\begin{array}{c|c} \mathbf{R}_2 & \mathbf{d}_2 \\ \hline \mathbf{0} & 1 \end{array}\right]$$

- t the time variable that parameterizes the entire motion, $t \in [0, \tau]$
- $E(t)$ the end effector frame at an arbitrary time t : $E(0)=E_1$, $E(\tau)=E_2$.
- $\mathbf{H}(t)$ the homogeneous transformation matrix for the displacement $D{:}E_1 \rightarrow E(t)$ with

$$\mathbf{H}(t) = \left[\begin{array}{c|c} \mathbf{R}(t) & \mathbf{d}(t) \\ \hline \mathbf{0} & 1 \end{array}\right]$$

Trajectory generation

We first consider the problem of generating a smooth trajectory between E_1 and E_2 without considering the geometry of the manipulator. The main idea is simple. If the object of interest is a point (for example, a reference point on the end effector) in Euclidean space R^3, it is meaningful to follow the shortest distance path. Thus the optimal trajectory is the straight line between the starting position and the goal position of the point. Unfortunately, the object of interest is a reference frame *E.* Thus we need the generalization of a straight line to a more complicated space (the space of all reference frames). In geometry such generalizations are called geodesics[2]. We will show how to generate such a shortest distance trajectory between the

[2] An airplane flying between two cities generally follows the geodesic between the two cities. The geodesic will be the shortest distance path and will appear curved. For example, an airplane flying from Tokyo to Los Angeles will fly north before turning south although Tokyo is north of Los Angeles. If the earth had been flat, the geodesic would simply be the straight line between the two cities.

start and goal position. The underlying mathematics is quite involved and can be found in a recent paper[3].

Let $\mathbf{H}(t)$ be the transformation matrix that relates the position and orientation of the end effector at time t with respect to the initial reference frame E_1. Note that E_1 is a fixed reference frame. The end effector frame is coincident with E_1 at t=0. Clearly, $\mathbf{H}(\tau)$ relates the goal position and orientation, E_2, with respect to E_1. Let $\mathbf{M} = \mathbf{H}(\tau)$. Since the start and goal frames are described by $\mathbf{Q}_1$ and $\mathbf{Q}_2$,

$$\mathbf{M} = (\mathbf{Q}_1)^{-1}\,\mathbf{Q}_2, \qquad \mathbf{M} = \left[\begin{array}{c|c} \mathbf{R}_1^T\mathbf{R}_2 & \mathbf{R}_1^T(\mathbf{d}_2 - \mathbf{d}_1) \\ \hline \mathbf{0} & 1 \end{array}\right] \tag{1}$$

where:

$$\mathbf{Q}_1 = \left[\begin{array}{c|c} \mathbf{R}_1 & \mathbf{d}_1 \\ \hline \mathbf{0} & 1 \end{array}\right], \qquad \mathbf{Q}_2 = \left[\begin{array}{c|c} \mathbf{R}_2 & \mathbf{d}_2 \\ \hline \mathbf{0} & 1 \end{array}\right]$$

Thus the trajectory of the end effector in reference frame E_1 is given by:

$$\mathbf{H}(t) = \left[\begin{array}{c|c} \mathbf{R}(t) & \mathbf{d}(t) \\ \hline \mathbf{0} & 1 \end{array}\right] \tag{2}$$

where the following boundary conditions must be satisfied:

$$\mathbf{H}(0) = \mathbf{I}_{4\times4}, \qquad \mathbf{R}(0)=\mathbf{I}_{3\times3}, \qquad \mathbf{d}(0)=\mathbf{0}_{3\times1} \tag{3}$$

$$\mathbf{H}(\tau) = \mathbf{M}, \qquad \mathbf{R}(\tau)=\mathbf{R}_1^T\mathbf{R}_2\,, \qquad \mathbf{d}(\tau)=\mathbf{R}_1^T(\mathbf{d}_2 - \mathbf{d}_1) \tag{4}$$

It turns out that this shortest distance trajectory[4] can be obtained by treating the rotation and translation parts of the trajectory independently. If we consider the rotation part of the trajectory, it is clear that the simplest trajectory from a given initial position, $\mathbf{R}(0)=\mathbf{I}$ to a new position, $\mathbf{R}(\tau)=\mathbf{R}_1^T\mathbf{R}_2$, is a uniform rotation of the end effector about a fixed axis. The existence of this axis is guaranteed by Euler's theorem. A simple solution for the translation part of this trajectory

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[3]M. Zefran and V. Kumar. On the generation of smooth trajectories for rigid bodies. Workshop on Computational Kinematics, Nice, September 4-6, 1995.

[4]It is clear that our discussion is quite superficial. We have not defined our measure of distance (a suitable metric) and therefore the notion of a shortest distance trajectory is not at all obvious. There is a natural way of defining a metric on the set of all rigid body displacements and it is this metric we refer to in this presentation.

is the uniform translation along a straight line from $\mathbf{d}(0)=\mathbf{0}$ to $\mathbf{d}(\tau)= \mathbf{R}_1^T(\mathbf{d}_2-\mathbf{d}_1)$ in three dimensional space. It turns out that these solutions do give the shortest distance trajectory (for a class of metrics) betwen E_1 and E_2.

Let the angle of rotation and the axis of rotation be given by φ and $\mathbf{u}$ respectively. From Equations (1.28) in the notes, we can easily derive the following formulae:

$$\varphi = \cos^{-1}\left[\frac{1}{2}\left(trace\left(\mathbf{R}_1^T\mathbf{R}_2\right)-1\right)\right] \tag{5}$$

$$\mathbf{U} = \frac{1}{2\sin\varphi}\left(\mathbf{R}_1^T\mathbf{R}_2-\mathbf{R}_2^T\mathbf{R}_1\right) \tag{6}$$

where $\mathbf{U}$ is the 3×3 skew symmetric matrix corresponding to the 3×1 vector $\mathbf{u}$. Note that there are many solutions for the angle of rotation[5] because the inverse cosine function is multivalued. If we find φ from (5) restricting the range of the inverse cosine function to the interval $[0,\pi]$, we can find the axis of rotation from (6) provided φ is not either 0 or π. If $\mathbf{R}_1^T\mathbf{R}_2 = \mathbf{I}$, $\varphi=0$ and $\mathbf{R}(\tau)=\mathbf{I}$. If the trace of the rotation matrix is -1, $\varphi=\pi$, and from the Euler-Lexell formula:

$$\mathbf{R}_1^T\mathbf{R}_2 = 2\mathbf{u}\mathbf{u}^T - \mathbf{I}$$

from which $\mathbf{u}$ can be solved.

We want the end effector to perform a uniform rotation of φ about $\mathbf{u}$ in τ seconds. Thus the velocity of rotation must be given by the angular velocity vector:

$$\omega = \frac{\varphi}{\tau}\,\mathbf{u}$$

and the rotation matrix is given by:

$$\mathbf{R}(t) = \mathbf{I}\cos\frac{\varphi t}{\tau} + \mathbf{u}\mathbf{u}^T\left(1-\cos\frac{\varphi t}{\tau}\right) + \mathbf{U}\sin\frac{\varphi t}{\tau} \tag{7}$$

5 For every rotation of angle φ about the axis $\mathbf{u}$, we can also obtain an equivalent axis-angle representation with a rotation $-\varphi$ about the axis $-\mathbf{u}$. Also for every solution $(\mathbf{u}, \varphi)$, we have other solutions $(\mathbf{u}, \varphi \pm 2k\pi)$ for all integer values of k.

Since the translation $\mathbf{R}_1^T(\mathbf{d}_2 - \mathbf{d}_1)$ must be uniform and performed in the same time interval, we get:

$$\mathbf{d}(t) = \mathbf{R}_1^T(\mathbf{d}_2 - \mathbf{d}_1)\frac{t}{\tau} \tag{8}$$

Equations (7) and (8) describe the displacement of the end effector from its initial position at E_1 to the final position E_2 in the fixed reference frame E_1.

Cartesian velocities

The end effector velocities can be obtained either in the end effector reference frame E (reference frame 6) or the fixed reference frame F (reference frame 0). We will determine the velocities in the end effector reference frame. The twist in the moving reference frame at any time t is given by:

$$^E\mathbf{t}(t) = \mathbf{H}(t)^{-1}\frac{d\mathbf{H}(t)}{dt} \tag{9}$$

A simple computation verifies that:

$$^E\Omega(t) = \mathbf{R}(t)^T\frac{d\mathbf{R}(t)}{dt} = \frac{\varphi}{\tau}\mathbf{U} \tag{10}$$

$$^E\mathbf{v}(t) = \mathbf{R}(t)^T\frac{d\mathbf{d}(t)}{dt} = \mathbf{R}(t)^T\frac{\mathbf{R}_1^T(\mathbf{d}_2 - \mathbf{d}_1)}{\tau} \tag{11}$$

The twist in the end effector frame can be transformed to any other reference frame. In the kth reference frame:

$$^k\mathbf{t} = \begin{bmatrix} ^k\mathbf{R}_6 & \mathbf{0} \\ ^k\mathbf{P}_6\,^k\mathbf{R}_6 & ^k\mathbf{R}_6 \end{bmatrix} {}^6\mathbf{t} = \begin{bmatrix} ^k\mathbf{R}_6 & \mathbf{0} \\ ^k\mathbf{P}_6\,^k\mathbf{R}_6 & ^k\mathbf{R}_6 \end{bmatrix}\begin{bmatrix} ^E\omega \\ ^E\mathbf{v} \end{bmatrix} \tag{12}$$

where $^k\mathbf{P}_6$ is the skew-symmetric matrix corresponding to $^k\mathbf{p}_6$ and

$$^k\mathbf{A}_6 = \left[\begin{array}{c|c} ^k\mathbf{R}_6 & ^k\mathbf{p}_6 \\ \hline \mathbf{0} & 1 \end{array}\right]$$

Joint velocities

To obtain the joint velocities for the desired end effector twist given by equations (10-11) or (12), one needs to premultiply the twist by the inverse of Jacobian. We have seen that letting k=3 allows us to develop analytical expressions for the joint rates. Once the desired joint rates are calculated, the joints can be commanded to move at the desired rates and the end effector moves in the desired trajectory.

Control algorithm

1. Given the starting and goal positions and orientations find the required displacement from Equation (1).
2. Compute the desired trajectory from Equations (7) and (8)
3. Find the expressions for the joint rates as analytical functions of the end effector velocities in a suitable reference frame (generally, frame 3 will work well).
4. For each time step:
   - 4.1 Compute the end effector velocity (10-11)
   - 4.2 Transform the velocity to frame 3
   - 4.3 Obtain joint rates from the analytical expressions in Step 3
   - 4.4 Send the desired joint rates to the motor controllers
   - 4.5 Wait for Δt seconds (sampling rate)
   - 4.6 Go to 4.1